

RA-14: Adaptive Lower Bounds and the Fixed-Accuracy Characterization

Abstract

In the exact two-sided matrix–vector oracle model, a self-contained argument proves

$$q_{\text{sp}}(n, k, \varepsilon) \geq \frac{k}{128} \log \frac{en}{k} \quad (n \geq 2, 1 \leq k < n, 0 < \varepsilon < 1/2).$$

Together with the preceding dimension-sensitive upper bound, this characterizes the complexity for every rank and dimension whenever accuracy is bounded away from zero. In particular, it closes the previously identified gap at $k = \lfloor n/\log n \rfloor$ and constant accuracy. The proof conditions on all but one direction of a random planted subspace, controls adaptive Gaussian likelihood ratios, and combines a ridge-dependent sphere covering with a rank-one determinant update.

A sharper version of the same information argument, a proved spectral-to-PCA postprocessor, and the finite-sample spectral theorem of Simchowitz–El Alaoui–Recht give matching $\Theta(k\varepsilon^{-1/2} \log(en/k))$ bounds under $n \geq C\varepsilon^{-3}(k + \log(1/\varepsilon))$ for a universal, unoptimized C . The imported result is a spectral concentration theorem, not an all-parameter query lower bound.

The full simultaneous finite-parameter RA-14 problem remains **PARTIAL**: neither theorem supplies matching bounds in all high-accuracy regimes. All new lower-bound arguments, the warm-start proof, and a condensed upper-bound proof appear below. No independent peer review, mathematical priority, or proof-assistant verification is claimed.

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1 Problem, results, and exact scope

The repository problem [1] asks for the worst-case query complexity, to universal constant factors, of the following task. An unknown $A \in \mathbb{R}^{n \times n}$ is accessible through either Ax or $A^\top x$; each individual vector product costs one query. All other exact-real computation is free. A randomized algorithm must return $Z \in \mathbb{R}^{n \times k}$ with $Z^\top Z = I_k$ and, for every fixed input A ,

$$\mathbb{P}\{\|A(I - ZZ^\top)\|_2 \leq (1 + \varepsilon)\sigma_{k+1}(A)\} \geq 0.99. \quad (1.1)$$

Here $n \geq 2$, $1 \leq k < n$, and $0 < \varepsilon < 1/2$. Algorithms, their random choices, and their outputs are measurable. The minimum worst-case integer query budget is $q_{\text{sp}}(n, k, \varepsilon)$. All logarithms are natural. A product with a block of b columns costs b queries, not one.

Theorem 1.1 (Universal lower bound and fixed-accuracy characterization). *For every admissible parameter triple,*

$$\max \left\{ k, \frac{k}{128} \log \frac{en}{k} \right\} \leq q_{\text{sp}}(n, k, \varepsilon) \leq \min \left\{ n, \left\lceil 12000 \frac{k}{\sqrt{\varepsilon}} \log \frac{en}{k} \right\rceil \right\}. \quad (1.2)$$

Consequently, for any fixed $\varepsilon_0 \in (0, 1/2)$, uniformly over n, k and $\varepsilon \in [\varepsilon_0, 1/2)$,

$$q_{\text{sp}}(n, k, \varepsilon) = \Theta_{\varepsilon_0} \left(k \log \frac{en}{k} \right).$$

In particular, the constants can be universal on the interval $1/16 \leq \varepsilon < 1/2$.

The k lower bound is proved in Section 2. The new logarithmic lower bound is proved in Sections 3–4. The upper bound is from the preceding continuation [3]; Appendix A gives its construction and proof. Notice that $k \log(en/k) \leq n$, by $\log x \leq x - 1$ for $x = n/k$.

Theorem 1.2 (Shrinking accuracy with a ratio-based dimension condition). *There is a universal constant $C > 0$ such that, for $0 < \varepsilon \leq 1/16$,*

$$n \geq C\varepsilon^{-3}(k + \log(1/\varepsilon)) \implies q_{\text{sp}}(n, k, \varepsilon) \geq \frac{1}{2048} \frac{k}{\sqrt{\varepsilon}} \log \frac{en}{k}. \quad (1.3)$$

The upper bound in (1.2) then matches to universal constant factors. The constant C is not optimized or numerically evaluated. The proof allows it to be very large; in particular it absorbs a fixed condition $\log(en/k) \geq 16384$ and the universal constant in the imported finite-sample spectral theorem.

The information-theoretic part of this theorem is proved here. Its only external probabilistic dependency is the explicitly stated spectral result in Section 6, from [4, Theorem 6.1].

1.1 A gap that is now closed

At $\varepsilon = 1/4$ and $k = \lfloor n/\log n \rfloor$, Theorem 1.1 gives

$$q_{\text{sp}}(n, k, 1/4) = \Theta \left(\frac{n \log \log n}{\log n} \right).$$

The previous package had only the corresponding upper bound and an $\Omega(n/\log n)$ global lower bound. The present argument supplies the missing $\log \log n$ factor against arbitrary adaptive algorithms, not just block-Krylov methods.

1.2 A gap that is not closed

For example, $k = 1$, $\varepsilon = 1/n$ violates (1.3) for large n . Neither the new theorems nor the previous dimension-restricted reductions produce a matching lower bound there. The upper bound is $O(\sqrt{n} \log n)$, and no assertion here identifies the exact order in that regime. A complete answer still requires further lower bounds or a sharper algorithm with matching bounds. The word *fixed* in “fixed-accuracy characterization” is essential.

2 The rank lower bound and oracle normalization

Lemma 2.1 (Unrevealed rectangular Gaussian block). *Let $H \in \mathbb{R}^{n \times k}$ have independent standard Gaussian entries. A deterministic adaptive procedure observes Hc and $H^\top x$. Let $C \subset \mathbb{R}^k$ and $X \subset \mathbb{R}^n$ be the spans of the respective query vectors. Conditional on the transcript,*

$$H = \bar{H} + P_{X^\perp} G P_{C^\perp}, \quad \bar{H} = P_X H + H P_C - P_X H P_C, \quad (2.1)$$

where $\bar{H}$ is known and the remaining block is a fresh standard Gaussian map from $C^\perp$ to $X^\perp$.

Proof. Given the transcript, the next vector is fixed. A new column query reveals one linear combination of the unrevealed columns. An orthogonal change of basis in $C^\perp$ makes it one column of an independent Gaussian matrix, leaving the other columns independent and unchanged. A new row query has the same argument in $X^\perp$. Redundant queries reveal nothing. Induction proves the claim, including for adaptive vectors. Inclusion–exclusion gives the displayed known part. $\square$

Proposition 2.2. $q_{\text{sp}}(n, k, \varepsilon) \geq k$. *On the promised class $\text{rank } A \leq k$, exactly k queries are necessary and sufficient.*

Proof. Use the nonsymmetric input $A = [H^\top; 0_{(n-k) \times n}]$. Its allowed queries are exactly those in Lemma 2.1. Write $r = \dim C$, $s = \dim X$. Before k total queries, $\text{rank}(H^\top X) = s$ almost surely: a new row response has a nondegenerate Gaussian component in $C^\perp$, of dimension $k - r > s$, so has probability zero of lying in the span of the s preceding row responses. Column queries do not change this rank assertion.

Fix the algorithm’s seed and a transcript after $q < k$ queries. Successful output $S = \text{range } Z$ must equal $\text{range } H$, because $\sigma_{k+1}(A) = 0$. If $X^\perp \not\subset S$, then for nonzero $c \in C^\perp$, the conditional affine Gaussian support of Hc is parallel to $X^\perp$. Its intersection with S has measure zero. If $X^\perp \subset S$, the row-response rank assertion gives $P_X \text{range } H = X$, whence $\text{range } H + X^\perp = \mathbb{R}^n$. Both spaces cannot lie in S , since $\dim S = k < n$. Success therefore has probability zero under this input law. Integrating over the random seed contradicts any pointwise 0.99 guarantee with fewer than k queries.

For sufficiency under the rank promise, query $A^\top$ on k independent Gaussian vectors. Their images span the row space almost surely. Orthonormalize and complete to dimension k when needed; the residual is zero. $\square$

Lemma 2.3 (Orthonormal queries and charged output extension). *For a symmetric input M , an algorithm with at most q queries can be simulated using at most q orthonormal query directions. After it returns a k -dimensional subspace, at most k additional queries suffice to make the cumulative query span contain that subspace. If a deterministic total budget $T \leq n$ is desired, the procedure can be padded with orthonormal dummy queries to have exactly T directions.*

Proof. The two oracle directions coincide on a symmetric input. Decompose each requested vector into its component in the already queried span and its orthogonal complement. The first product is known by linearity; query the normalized second component only if it is nonzero. This reproduces the original transcript. Apply the same orthogonalization to the output columns, making at most k further queries. Fill unused directions by a deterministic measurable orthonormal completion. The simulation, extension, and padding define a procedure under every input law used below. $\square$

The output-extension queries are included in every lower-bound budget. The proof does not grant free access to MZ .

3 Adaptive Gaussian likelihood and spherical information

Consider the planted family

$$M = W + \lambda U U^\top, \quad (3.1)$$

where U is uniform in $\text{Stief}(n, k)$ and independent of W . We use the normalization $W_{ij} \sim N(0, 1/n)$ independently above the diagonal, $W_{ii} \sim N(0, 2/n)$ independently on it, and $W_{ji} = W_{ij}$. This is denoted $\text{GOE}(n)$.

Fix the algorithm's seed and use orthonormal query directions $v_1, \dots, v_T$. Define

$$V_t = [v_1, \dots, v_t], \quad P_t = V_t V_t^\top, \quad A_t = n U^\top P_t U \quad (0 \leq t \leq T).$$

Thus $A_0 = 0$, $0 \preceq A_t \preceq n I_k$, and each increment $A_{t+1} - A_t$ is positive semidefinite of rank at most one. These facts hold pathwise.

The conditioning and determinant framework is related to [4, Sections 4–5]. The estimates used here, especially the ridge-dependent covering bound, are proved below rather than imported from a dimension-restricted query theorem.

3.1 Conditioning on all but one planted direction

Fix a deterministic $e \in \mathbb{R}^k$ with $\|e\|_2 = 1$, and choose an orthogonal matrix in column coordinates whose first column is e . In these coordinates write $U = [u, B]$. Conditional on B , $u = Ue$ is uniform on the unit sphere in $B^\perp$, whose dimension is

$$D = n - k + 1.$$

The actual input is $M = W + \lambda B B^\top + \lambda u u^\top$. The reference input is

$$M_0 = W + \lambda B B^\top. \quad (3.2)$$

The algorithm is not given B . Under the reference law, u is independent of its transcript conditional on B . Each fixed e uses its own valid reference law; the resulting bounds are all probabilities under the same actual law and may subsequently be union-bounded over deterministic e 's.

Put $S_t = n \|V_t^\top u\|_2^2 = e^\top A_t e$. The next vector v_{t+1} is a function of the first t replies. Let L_t be the likelihood ratio of the first t replies under the actual law relative to the reference law, with u, B fixed.

Lemma 3.1 (Truncated likelihood moment). *For $\eta > 0$ and $a \geq 0$,*

$$\mathbb{E}_0[L_t^{1+\eta} \mathbf{1}_{\{S_t \leq a\}} \mid u, B] \leq \exp\left(\frac{\eta(1+\eta)\lambda^2 a}{2}\right). \quad (3.3)$$

This remains valid for fully adaptive orthonormal query directions.

Proof. At step i , the components of Mv_i in range V_{i-1} are already known by symmetry. It is enough to observe $y_i = (I - P_{i-1})Mv_i$. Conditional on the preceding transcript and fixed signal parameters, the unrevealed GOE block on range $(I - P_{i-1})$ is independent of the revealed block. This follows inductively: given the past, the next direction is fixed; a rotation within the unrevealed space makes it the first coordinate, and revealing its row and column leaves an independent GOE block on the orthogonal complement. The variance normalization remains $1/n$.

Consequently, on this complement the response covariance, the same under both laws, is

$$\Sigma_i = \frac{1}{n}((I - P_{i-1}) + v_i v_i^\top).$$

The difference of conditional means is

$$\mu_i = \lambda \langle u, v_i \rangle (I - P_{i-1})u.$$

Using an orthonormal coordinate basis for the complement, Σ_i is invertible, and

$$\begin{aligned} h_i &:= \mu_i^\top \Sigma_i^{-1} \mu_i = n\lambda^2 \langle u, v_i \rangle^2 \left(\|(I - P_{i-1})u\|_2^2 - \frac{1}{2} \langle u, v_i \rangle^2 \right) \\ &\leq n\lambda^2 \langle u, v_i \rangle^2. \end{aligned}$$

Under the reference law, the log-likelihood increment is $\xi_i - h_i/2$, where conditionally ξ_i is centered Gaussian of variance h_i . Both h_i and the query direction are predictable given u, B and the preceding transcript. Write $p = 1 + \eta$. The random variable

$$K_t = \exp \left(p \sum_{i=1}^t \xi_i - \frac{p^2}{2} \sum_{i=1}^t h_i \right)$$

is a mean-one exponential martingale, by the conditional Gaussian moment-generating identity. The finite-step conditional expectations are legitimate; in particular $0 \leq h_i \leq n\lambda^2$. Since

$$L_t^p = K_t \exp \left(\frac{\eta(1+\eta)}{2} \sum_{i=1}^t h_i \right), \quad \sum_{i=1}^t h_i \leq \lambda^2 S_t,$$

truncation to $S_t \leq a$ and $\mathbb{E}_0 K_t = 1$ prove (3.3). $\square$

Lemma 3.2 (A spherical moment bound). *Let u be uniform on the unit sphere of a fixed D -dimensional subspace. For any fixed ambient orthonormal p -frame V and $0 < \theta < 1/2$,*

$$\mathbb{E} \exp(\theta D \|V^\top u\|_2^2) \leq (1 - 2\theta)^{-p/2}. \quad (3.4)$$

Proof. Compress $VV^\top$ to the D -dimensional subspace. It is positive semidefinite, has eigenvalues at most one, and rank at most p . If $p < D$, its quadratic form is pointwise dominated, in an eigenbasis, by the sum X of the first p squared coordinates of a uniform unit vector in $\mathbb{R}^D$. Writing that vector as a normalized standard Gaussian vector gives

$$\mathbb{E} X^j = \frac{(p/2)_j}{(D/2)_j}, \quad \mathbb{E} (DX)^j \leq 2^j (p/2)_j \quad (j \geq 0),$$

where $(a)_j = a(a+1) \cdots (a+j-1)$. Expanding the exponential with nonnegative terms gives (3.4) by the binomial series. If $p \geq D$, the quadratic form is at most one, and $e^{\theta D} \leq (1 - 2\theta)^{-p/2}$, using $-\log(1 - 2\theta) \geq 2\theta$. $\square$

Proposition 3.3 (One-step information inequality). *Let $c = n/(n - k + 1)$. For $a, b \geq 0$, $\eta > 0$, and $0 < \theta < 1/2$,*

$$\mathbb{P}\{S_t \leq a, S_{t+1} > b\} \leq \exp \left\{ \frac{\eta \lambda^2 a}{2} + \frac{\eta}{1 + \eta} \left(-\frac{\theta b}{c} + \frac{t+1}{2} \log \frac{1}{1 - 2\theta} \right) \right\}. \quad (3.5)$$

In particular, if $k \leq n/2$, then

$$\mathbb{P}\{S_t \leq a, S_{t+1} > b\} \leq \exp \left\{ \frac{\lambda^2 a}{2} + \frac{t+1}{4} - \frac{b}{16} \right\}. \quad (3.6)$$

For $s > 0$, taking $\eta = s$ and $\theta = 1/[2(1 + s)]$ instead yields

$$\mathbb{P}\{S_t \leq a, S_{t+1} > b\} \leq \exp \left\{ \frac{s}{2} \left[\lambda^2 a - \frac{b}{c(1 + s)^2} + \frac{t+1}{1 + s} \log(1 + 1/s) \right] \right\}. \quad (3.7)$$

Proof. Conditional on B , consider the joint reference law of the transcript and the independent uniform u . The event $S_{t+1} > b$ is determined by u and the first t replies, since the next vector is predictable. Under the reference law its probability is at most

$$\exp \left(-\frac{\theta b}{c} + \frac{t+1}{2} \log \frac{1}{1 - 2\theta} \right)$$

by Lemma 3.2 and Markov's inequality. Hölder's inequality, with exponents $1 + \eta$ and $(1 + \eta)/\eta$, applied to $L_t \mathbf{1}_{\{S_t \leq a\}} \mathbf{1}_{\{S_{t+1} > b\}}$, and then Lemma 3.1, gives (3.5). It is uniform in B , so conditioning may be removed. For (3.6), use $c \leq 2$, $\eta = 1$, $\theta = 1/4$, and $\log 2 < 1$. The second specialization is direct substitution. $\square$

4 Matrix growth and the universal logarithmic lower bound

The ridge parameter Δ is retained in the sphere covering radius. Dropping it would replace the useful $\log(n/k)$ dependence by a cruder dimension logarithm.

Lemma 4.1 (Constant-strength matrix growth). *Suppose $k \leq n/2$, $\lambda \geq 2$, and $1 \leq T \leq \Delta \leq n$. Put $\rho = 16\lambda^2$ and $R = 2\rho = 32\lambda^2$. Except with probability at most*

$$3T \left(1 + \frac{16n}{\Delta} \right)^k e^{-\lambda^2 \Delta/3}, \quad (4.1)$$

all $0 \leq t < T$ satisfy

$$A_{t+1} + \Delta I \preceq R(A_t + \Delta I). \quad (4.2)$$

On that event,

$$\det(A_T + \Delta I) \leq R^T \Delta^k. \quad (4.3)$$

Proof. Fix a unit e and a step t . A violation $S_{t+1} + \Delta > \rho(S_t + \Delta)$, on $S_t \in [j, j + 1)$, implies $S_{t+1} > \rho j + (\rho - 1)\Delta$. By (3.6) and a sum over $j \geq 0$, its probability is at most

$$\frac{\exp\{\lambda^2/2 + (t+1)/4 - (\rho - 1)\Delta/16\}}{1 - e^{-\lambda^2/2}} \leq 3e^{-\lambda^2 \Delta/3}.$$

For the last step use $t+1 \leq \Delta$, $\Delta \geq 1$, $\lambda^2 \geq 4$: the numerator exponent is at most $-(\lambda^2/2 - 5/16)\Delta \leq -\lambda^2\Delta/3$ and the reciprocal denominator is less than 3.

Take a deterministic Euclidean δ -net of the unit sphere of $\mathbb{R}^k$, where $\delta = \Delta/(8n)$. A maximal separated set and the disjoint-ball volume argument give cardinality at most $(1 + 2/\delta)^k = (1 + 16n/\Delta)^k$. If (4.2) fails at a unit e , then

$$e^\top (A_{t+1} - \rho A_t) e - (\rho - 1)\Delta > \rho\Delta.$$

Replacing e by a net vector f changes the quadratic form by at most

$$2 \|A_{t+1} - \rho A_t\|_2 \|e - f\|_2 \leq 2n(1 + \rho)\delta \leq \rho\Delta/2.$$

Thus the weaker factor- ρ inequality fails at f . Union-bounding over the net and T steps proves (4.1).

For the determinant assertion, put $B_t = A_t + \Delta I \succ 0$ and write $B_{t+1} = B_t + ww^\top$. From (4.2), $I + B_t^{-1/2}ww^\top B_t^{-1/2} \preceq RI$. This is a rank-one update of the identity. Its determinant is its one potentially nonunit eigenvalue, at most R . Therefore $\det B_{t+1} \leq R \det B_t$; iterate from $B_0 = \Delta I$. The exponent is T , not kT . $\square$

Lemma 4.2 (A crude GOE norm bound). *For the stated normalization,*

$$\mathbb{P}\{\|W\|_2 > 10\} \leq 2 \exp\{n(\log 9 - 25/4)\} \leq 2e^{-4n} < 0.001 \quad (n \geq 2).$$

Proof. A $1/4$ -net of the unit sphere in $\mathbb{R}^n$ has cardinality at most 9^n . For a symmetric W , approximation of a maximizing unit vector gives $\|W\|_2 \leq 2 \max_{x \text{ in net}} |x^\top Wx|$. For fixed unit x , $x^\top Wx \sim N(0, 2/n)$. A two-sided Gaussian tail at 5 and the union bound prove the first inequality. Finally $25/4 - \log 9 > 4$. $\square$

Proof of the lower bound in Theorem 1.1. Probabilities in this proof include the algorithm's random seed. The preceding information bounds hold for each seed and therefore also after averaging. Set $x = n/k$ and $L = \log(ex)$. If $L \leq 64$, Proposition 2.2 gives $q_{\text{sp}} \geq k \geq kL/128$. Hence assume $L > 64$ and, for contradiction, that an algorithm uses $q < kL/128$ queries. Use (3.1) with $\lambda = 100$. Apply Lemma 2.3, extending the query span to include the output and padding to

$$T = q + k \leq k(1 + L/128), \quad \Delta = 2kL.$$

Here $k \leq n/2$, $T \leq \Delta \leq n$, because $x = e^{L-1}$ and $L > 64$.

On $\|W\|_2 \leq 10$, the rank- k perturbation implies $\sigma_{k+1}(M) \leq 10$. One proof is to restrict M to $\text{range}(U)^\perp$ in the singular-value min-max principle. On a successful run, (1.1) therefore gives $\|M(I - ZZ^\top)\|_2 \leq 15$. The triangle inequality yields

$$100 \|U^\top (I - ZZ^\top)\|_2 \leq \|M(I - ZZ^\top)\|_2 + \|W(I - ZZ^\top)\|_2 \leq 25.$$

Hence $U^\top ZZ^\top U \succeq (15/16)I_k$. The extended query span contains $\text{range } Z$, so

$$A_T \succeq \frac{15n}{16} I_k. \tag{4.4}$$

The joint event of success and $\|W\|_2 \leq 10$ has probability at least 0.989: the guarantee is pointwise in M and can be integrated over the planted input law.

Lemma 4.1 has $R = 320000$. Its exceptional probability is less than 0.001. Indeed,

$$\log(3T) \leq 2kL, \quad k \log(1 + 16n/\Delta) \leq 2kL, \quad \lambda^2\Delta/3 = (20000/3)kL.$$

For the first inequality use $T \leq n$ and $k \log(en/k) \geq 1 + \log n$; for the second use $\Delta = 2kL$, $L > 64$, and $x > 1$.

If both (4.4) and the determinant event hold, then

$$k \log \left(1 + \frac{15x}{32L} \right) \leq T \log 320000. \quad (4.5)$$

The logarithm on the left is at least $L/2$ for $L \geq 64$: it is at least $L - 1 + \log(15/32) - \log L$, and this exceeds $L/2$ at 64 and has positive difference derivative thereafter. But, since $\log 320000 < 16$,

$$T \log 320000 < 16k(1 + L/128) \leq \frac{3}{8}kL.$$

This contradicts (4.5). The two required events have a strictly positive intersection probability, so the contradiction rules out the proposed algorithm. The argument was uniform in the algorithm's fixed seed; averaging restores arbitrary algorithmic randomness. Together with the upper bound, this proves Theorem 1.1. $\square$

5 A sharper information bound near the critical spike

For shrinking accuracy, the spike strength must approach one rather than remain 100. The coarser information bound loses that dependence. The following lemma retains it.

Lemma 5.1 (Near-critical matrix growth). *Let $0 < s \leq 1$, $\lambda = 1 + s$, and $c = n/(n - k + 1) \leq 1 + s$. Write $\ell = \log(1 + 1/s)$ and assume*

$$1 \leq T \leq n, \quad \max\{T\ell/s, 2/s^3\} \leq \Delta \leq n. \quad (5.1)$$

Except with probability at most

$$2T \left(1 + \frac{32n}{s\Delta} \right)^k e^{-s^2\Delta}, \quad (5.2)$$

all steps satisfy $A_{t+1} + \Delta I \preceq (1 + s)^8(A_t + \Delta I)$, and

$$\det(A_T + \Delta I) \leq (1 + s)^{8T} \Delta^k. \quad (5.3)$$

Proof. Put $\rho = (1 + s)^7$ and fix a unit e . Bin $S_t + \Delta$ into $[z_j, (1 + s)z_j)$, where $z_j = (1 + s)^j \Delta$, $j \geq 0$. If $S_{t+1} + \Delta > \rho(S_t + \Delta)$ in that bin, apply (3.7) with

$$a = (1 + s)z_j - \Delta, \quad b = \rho z_j - \Delta.$$

Since $1 \leq c \leq 1 + s$, the coefficient of z_j inside the square bracket of (3.7) is at most

$$(1 + s)^3 - \frac{(1 + s)^7}{(1 + s)^3} = -s(1 + s)^3.$$

The coefficient of Δ is at most $-(1 + s)^2 + (1 + s)^{-2} \leq -2s$. Also $(t + 1)\ell/(1 + s) \leq T\ell \leq s\Delta$. The exponent is therefore at most

$$-\frac{s^2}{2}(z_j + \Delta).$$

By $(1+s)^j \geq 1+js$, summing over j bounds the weak-growth failure probability at this step by

$$\frac{e^{-s^2\Delta}}{1 - e^{-s^3\Delta/2}} \leq 2e^{-s^2\Delta},$$

using $s^3\Delta/2 \geq 1$.

Choose a deterministic $s\Delta/(16n)$ -net of the unit sphere of $\mathbb{R}^k$, of cardinality at most $(1 + 32n/(s\Delta))^k$. A violation of the stronger factor $(1+s)\rho$ produces an excess larger than $s\rho\Delta$ over the weak factor- ρ inequality. Moving to a net vector changes that excess by at most

$$2n(1+\rho)\frac{s\Delta}{16n} \leq s\rho\Delta/4.$$

Thus a stronger violation anywhere implies a weak violation on the net. Union-bound over the net and all T steps. The same rank-one determinant argument as in Lemma 4.1 proves (5.3). $\square$

6 The finite-sample spectral input

This section is the only place the shrinking-accuracy theorem imports a non-elementary probability result. The all-parameter constant-accuracy lower bound above does not depend on it.

For $\lambda > 1$, define

$$L_0 = \lambda + \lambda^{-1}, \quad g = \frac{(\lambda - 1)^2}{\lambda^2 + 1}, \quad gL_0 = L_0 - 2. \quad (6.1)$$

We use the following statement from Simchowitz–El Alaoui–Recht [4, Theorem 6.1], with its precision parameter renamed ξ to distinguish it from RA-14 accuracy.

Theorem 6.1 (Imported finite-sample deformed-Wigner spectrum). *There is a universal $C_0 > 0$ such that the following holds for the model (3.1). Let $0 < \kappa \leq 1/2$, $0 < \xi \leq g \min\{1/2, 1/(\lambda^2 - 1)\}$, and $\delta > 0$. If*

$$n \geq C_0 \left[\frac{k + \log(1/\delta)}{g\xi^2} + (\kappa g)^{-3} \log \frac{1}{\kappa g} \right], \quad (6.2)$$

then, with probability at least $1 - 9\delta$,

$$\|W\|_2 \leq 2 + \kappa g L_0, \quad [\lambda_k(M), \lambda_1(M)] \subset L_0[1 - \xi, 1 + \xi]. \quad (6.3)$$

No proof of this imported spectral concentration theorem is claimed here. The source is the 2020 arXiv v2 version of the STOC 2018 paper; the exact theorem number and all dimension and probability qualifiers are retained.

For $0 < \varepsilon \leq 1/16$, choose

$$s = 4\sqrt{\varepsilon} \leq 1, \quad \lambda = 1 + s, \quad \kappa = 1/100, \quad \xi = g/100, \quad \delta = 1/900. \quad (6.4)$$

Then $g = s^2/(\lambda^2 + 1)$ lies between $s^2/5$ and $s^2/2$, and $g \geq 16\varepsilon/5$. The precision restriction holds because $\lambda \leq 2$. A sufficient condition for (6.2) is

$$n \geq C_R s^{-6} (k + \log(e/s)) \quad (6.5)$$

for a universal C_R . Indeed $g\xi^2 = g^3/10^4$, $(\kappa g)^{-3} = 10^6 g^{-3}$, and $\log(100/g) = O(\log(e/s))$. Under (6.5), define $\mathcal{E}$ to be the event

$$w := \|W\|_2 \leq 2 + gL_0/100, \quad (1 - g/100)L_0 \leq \lambda_k(M) \leq \lambda_1(M) \leq (1 + g/100)L_0. \quad (6.6)$$

Then $\mathbb{P}(\mathcal{E}) \geq 0.99$.

On $\mathcal{E}$, a positive rank- k perturbation and eigenvalue interlacing give $|\lambda_j(M)| \leq w$ for all $j > k$. Also

$$\lambda_k(M) - w \geq 0.98gL_0 > 0.$$

Thus the top k algebraic eigenvalues are positive and also the top k singular values. Writing $a = \lambda_k(M)$ and $b = \sigma_{k+1}(M) \leq w$, we have

$$\frac{a-b}{a} \geq \frac{0.98g}{1+g/100} \geq 0.97g.$$

When $b > 0$, it follows that $a/b \geq 1 + 0.97g > 1 + \frac{3}{2}\varepsilon$. The $b = 0$ case is harmless. These are exactly the spectral hypotheses needed by the next postprocessing theorem.

7 Spectral approximation to a PCA trace objective

For completeness, the warm-start result from the first package [2] is proved here. A valid spectral approximation alone need not have a large PCA trace, so this reduction cannot be skipped.

Theorem 7.1 (Deterministic warm-start postprocessor). *Let M be symmetric, with top k eigenvalues positive and at least a , and remaining eigenvalues of magnitude at most b . Suppose*

$$a \geq (1 + \frac{3}{2}\varepsilon)b, \quad Z^\top Z = I_k, \quad \|M(I - ZZ^\top)\|_2 \leq (1 + \varepsilon)b,$$

where $0 < \varepsilon < 1/2$. A deterministic algorithm, not knowing a or b , uses at most $k\lceil 10/\sqrt{\varepsilon} \rceil$ additional products with M and returns orthonormal $\hat{V}$ such that

$$\text{tr}(\hat{V}^\top M \hat{V}) \geq (1 - \varepsilon/200) \sum_{i=1}^k \lambda_i(M). \quad (7.1)$$

Proof. Assume $b > 0$, write $\tau = (1 + \varepsilon)b$, and use an eigenbasis $M = \text{diag}(\Lambda_1, \Lambda_2)$, with entries $\lambda_i \geq a$ and $|\mu_j| \leq b$. If V denotes the top eigenbasis, then $V^\top = \Lambda_1^{-1}V^\top M$, so

$$\|V^\top(I - ZZ^\top)\|_2 \leq \tau/a < 1.$$

Hence $V^\top Z$ is invertible and the output range is a graph range $[I_k; F_0]$. Applying the residual condition to the perpendicular vectors $(-F_0^\top y, y)$ gives

$$F_0(\Lambda_1^2 - \tau^2 I)F_0^\top \preceq \tau^2 I - \Lambda_2^2. \quad (7.2)$$

The two diagonal matrices are positive definite. Therefore

$$F_0 = (\tau^2 I - \Lambda_2^2)^{1/2} K (\Lambda_1^2 - \tau^2 I)^{-1/2}, \quad \|K\|_2 \leq 1, \quad \|K\|_F^2 \leq k.$$

For $m \geq 1$, define the degree- $(m-1)$ polynomial

$$p_m(u) = \frac{T_m(u) - 1}{u - 1}, \quad p_m(1) = m^2,$$

where T_m is the Chebyshev polynomial. The identity $p_m(u) = m + 2 \sum_{j=1}^{m-1} (m-j)T_j(u)$ proves that p_m is positive and nondecreasing on $[1, \infty)$. For $-1 \leq u \leq 1$,

$$0 \leq p_m(u) \leq \min\{m^2, 2/(1-u)\}, \quad (7.3)$$

with the second bound omitted at $u = 1$. This follows from $p_m(\cos \theta) = (\sin(m\theta/2)/\sin(\theta/2))^2$ and $|T_m(u) - 1| \leq 2$. For $u > 1$, $p_m(u) = (\cosh(m \operatorname{arcosh} u) - 1)/(u - 1)$.

The range of $p_m(M/b)Z$ is the graph $[I; F]$, where $F = p_m(\Lambda_2/b)F_0p_m(\Lambda_1/b)^{-1}$. An orthonormal basis is $Y = [I; F](I + F^\top F)^{-1/2}$. Put $C = (I + F^\top F)^{-1}$. Without any commutation assumption on Λ_1 and $F^\top F$, positivity gives

$$\begin{aligned} \sum_i \lambda_i - \operatorname{tr}(Y^\top MY) &= \operatorname{tr}((\Lambda_1 - aI)F^\top FC) + \operatorname{tr}(F^\top (aI - \Lambda_2)FC) \\ &\leq \sum_{i,j} (\lambda_i - \mu_j) |F_{ji}|^2. \end{aligned} \quad (7.4)$$

Indeed $0 \preceq F^\top FC \preceq F^\top F$, $0 \prec C \preceq I$, and both matrices paired with these in the traces are positive semidefinite. Substitution from (7.2) bounds (7.4) by k times the supremum of

$$B(\lambda, \mu) = \frac{(\lambda - \mu)(\tau^2 - \mu^2)}{\lambda^2 - \tau^2} \left(\frac{p_m(\mu/b)}{p_m(\lambda/b)} \right)^2. \quad (7.5)$$

For fixed μ , $(\lambda - \mu)/(\lambda^2 - \tau^2)$ decreases on $\lambda > \tau$, since its derivative numerator is $\mu^2 - \tau^2 - (\lambda - \mu)^2 < 0$. The polynomial denominator increases. Thus the worst leading value can be replaced by $a_0 = (1 + 3\varepsilon/2)b$.

Choose $m = \lceil 10/\sqrt{\varepsilon} \rceil$ and scale $b = 1$. Set $e = \varepsilon$, $t_0 = 1 + e$, and $v = 1 - \mu \in [0, 2]$. The numerator of (7.5), before its leading-value denominator, is

$$N = p_m(\mu)^2(3e/2 + v)(e + v)(2 + e - v).$$

When $v \geq e$, (7.3) gives $N \leq 50$. When $v \leq e$, it gives $N \leq (25/2)m^4e^2 < (25/2)11^4 < 200000$. Also $a_0^2 - t_0^2 \geq e$, and $\operatorname{arcosh}(1 + 3e/2) \geq \sqrt{e}$, since $\cosh x \leq 1 + x^2$ for $0 \leq x \leq 1$. Hence

$$p_m(a_0)^2(a_0^2 - t_0^2) \geq \frac{(\cosh 10 - 1)^2}{(9/4)e} > \frac{4 \cdot 10^8}{9e}.$$

The exact rational certificate $\sum_{j=1}^7 10^{2j}/(2j)! = 5860808350/567567 > 10000$ justifies the last strict bound. Restoring b , we obtain $B(\lambda, \mu) < (9/2000)\varepsilon b < \varepsilon b/200$. Thus $\operatorname{tr}(Y^\top MY) \geq \sum_i \lambda_i - k\varepsilon b/200 \geq (1 - \varepsilon/200) \sum_i \lambda_i$.

Finally $p_m(M/b)Z$ lies in the block-Krylov space $\operatorname{range}[Z, MZ, \dots, M^{m-1}Z]$. Build this space and return its top k algebraic Ritz vectors. The variational principle gives at least the trace of Y . Orthonormalize successive frontier blocks, each of width at most k , and retain their images under M for the compressed matrix. Processing the m frontiers, including the last image needed for compression, costs at most km products. If the space becomes invariant or fills the ambient space, stop earlier. This algorithm never needs b or evaluates p_m ; the polynomial is only an existence certificate within the computed space. If $b = 0$, the starting subspace contains range M already and the same invariant-space argument applies. $\square$

Lemma 7.2 (PCA trace forces many planted directions). *Under the event $\mathcal{E}$ in (6.6), an orthonormal $\hat{V}$ satisfying (7.1) obeys*

$$\lambda_{\lceil k/2 \rceil}(\hat{V}^\top U U^\top \hat{V}) \geq \beta := \frac{gL_0}{4\lambda} = \frac{s^2}{4\lambda^2} \geq \frac{s^2}{16}. \quad (7.6)$$

Proof. Let $r' = \lceil k/2 \rceil$ and $C = \widehat{V}^\top U U^\top \widehat{V}$. If $\lambda_{r'}(C) < \beta$, at least $m = k - r' + 1 \geq k/2$ eigenvalues of C are below β . Since $\widehat{V}^\top M \widehat{V} \preceq wI + \lambda C$, its corresponding m smaller eigenvalues are at most

$$w + \lambda\beta \leq L_0(1 - 0.99g) + gL_0/4 = L_0(1 - 0.74g).$$

All its eigenvalues are at most $\lambda_1(M) \leq L_0(1 + 0.01g)$. Thus its trace is at most $kL_0(1 + 0.01g) - 0.75mgL_0 \leq kL_0(1 - 0.365g)$. But (7.1) and (6.6) give a trace at least $kL_0(1 - g/100)(1 - \varepsilon/200) \geq kL_0(1 - 0.012g)$, using $\varepsilon/200 \leq g/640$. This is a contradiction. $\square$

8 Proof of the shrinking-accuracy theorem

We now combine the information, spectral, and postprocessing components, with all query costs retained.

Proof of Theorem 1.2. Probabilities again include the algorithm's seed, with the information bound averaged over seeds. Use the parameters in (6.4), put $x = n/k$, $L = \log(ex)$, and $\ell = \log(1 + 1/s)$. Enlarge the universal constant C in (1.3) so that its hypothesis implies all of

$$n \geq C_R s^{-6}(k + \log(e/s)), \quad x \geq s^{-6}, \quad L \geq 16384, \quad c \leq 1 + s. \quad (8.1)$$

This is possible: $\varepsilon^{-3} = 4096s^{-6}$, $\log(1/\varepsilon) = \log 16 + 2\log(1/s)$, and a sufficiently large fixed lower bound on x/s^{-6} ensures the last two conditions. For example $x \geq 2s^{-6}$ gives $k/n \leq s/2$, which implies $c \leq 1 + s$ for $s \leq 1$. A fixed factor e^{16384} suffices for the stated logarithmic threshold. No particular numerical value for the imported C_R is assumed.

Suppose, for contradiction, that $q \leq kL/(512s)$ queries suffice. On the symmetric planted input, run this algorithm, append the postprocessor from Theorem 7.1, then append queries spanning its output. The total before dummy padding is at most

$$T := q + k\lceil 10/\sqrt{\varepsilon} \rceil + k \leq \frac{k}{s} \left(\frac{L}{512} + 42 \right), \quad (8.2)$$

because $10/\sqrt{\varepsilon} = 40/s$ and $s \leq 1$. Here T on the left is an integer, and the expression on the right is a bound, not a rounded budget. Set

$$\Delta = \frac{100kL\ell}{s^2} + \frac{2}{s^3}. \quad (8.3)$$

Clearly $\Delta \geq T\ell/s$ and $\Delta \geq 2/s^3$, since $100L \geq L/512 + 42$.

We next verify $\Delta \leq n$ and the determinant scale quantitatively. With $\beta \geq s^2/16$ from (7.6), the inequality $a + b \leq 2\max\{a, b\}$ gives

$$\frac{n\beta}{\Delta} \geq \min \left\{ \frac{xs^4}{3200L\ell}, \frac{kxs^5}{64} \right\}. \quad (8.4)$$

Write $u = \log(1/s)$. By $x \geq s^{-6}$, $u \leq (L - 1)/6$, while $\ell \leq u + \log 2 \leq L$. Therefore the logarithm of the first expression in (8.4) is at least

$$\frac{L - 1}{3} - \log 3200 - 2\log L \geq L/4,$$

and that of the second is at least

$$\frac{L-1}{6} - \log 64 \geq L/8.$$

Both last inequalities hold at $L = 16384$ and their differences are increasing thereafter. We have proved

$$\log(n\beta/\Delta) \geq L/8. \quad (8.5)$$

Since $\beta \leq 1$, this also proves $\Delta \leq n$. Moreover $\ell > 1/2$ gives $\Delta \geq 50kL/s^2 \geq T$ directly from (8.2); hence $T \leq n$. The padded procedure is well defined and Lemma 5.1 applies.

Its exceptional probability is less than 0.01. Indeed $\ell > 1/2$ gives $s^2\Delta \geq 50kL$, while

$$\frac{32n}{s\Delta} \leq \frac{32xs}{50L} \leq x, \quad k \log(1 + 32n/(s\Delta)) \leq kL, \quad \log(2T) \leq \log(2n) \leq kL.$$

The logarithm of (5.2) is therefore at most $-48kL$.

The event $\mathcal{E}$ has probability at least 0.99. The original algorithm succeeds with probability at least 0.99 for every input, so success and $\mathcal{E}$ have joint probability at least 0.98. On that joint event, the verified gap from Section 6 makes the postprocessor valid, and Lemma 7.2 supplies at least $k/2$ planted overlap eigenvalues of size β . The two $k \times k$ Gram matrices $\hat{V}^\top U U^\top \hat{V}$ and $U^\top \hat{V} \hat{V}^\top U$ have the same eigenvalues. Appending the output to the query span therefore implies

$$\det(A_T + \Delta I) \geq \Delta^k (1 + n\beta/\Delta)^{k/2}.$$

Intersecting with the matrix-growth event still has positive probability. On this intersection, (5.3) and (8.5) give

$$8T \log(1+s) \geq \frac{k}{2} \log(1 + n\beta/\Delta) \geq \frac{kL}{16},$$

so $T \geq kL/(128s)$, since $\log(1+s) \leq s$. This contradicts (8.2), because $L/512 + 42 < L/128$ whenever $L \geq 16384$. Thus $q > kL/(512s) = kL/(2048\sqrt{\varepsilon})$, proving (1.3). All probability estimates were uniform in the fixed algorithmic seed, so randomized adaptive algorithms are included. $\square$

9 Consequences, limitations, and proof dependencies

The first theorem answers the full rank–dimension dependence at constant accuracy. The second gives a sufficient dimension condition linear in k , in place of the unspecified power of k in the preceding headline theorem. For example, if $k = \lfloor n^{0.9} \rfloor$ and $\varepsilon = n^{-0.02}$, its dimension condition holds eventually, and

$$q_{\text{sp}}(n, k, \varepsilon) = \Theta(n^{0.91} \log n).$$

Similarly, $k = \lfloor n/\log n \rfloor$ and $\varepsilon = (\log n)^{-1/4}$ eventually give $\Theta(n \log \log n / (\log n)^{7/8})$. These are asymptotic consequences, not estimates of the very conservative finite constants.

The dimension condition in Theorem 1.2 is a sufficient condition proved here, not a necessary threshold for the problem. It must not be removed from the lower bound. In particular the exact n -query recovery algorithm supplies only an upper cap. It does not establish an n lower bound outside the spectral construction's range.

The earlier claimed large-dimension lower bound and the Bakshi–Narayanan rank-one padding transfer remain in the unchanged prior archive, with their original hypotheses [2, 6]. They are not silently strengthened here. A known block-diagonal construction would let a single vector query probe every block in parallel, so it cannot replace the conditional planted-subspace argument above. A lower bound on Krylov iteration depth likewise does not by itself cover all adaptive vector-query algorithms.

Claim		Exact dependency and scope
$q \geq k$		Gaussian completion; two-sided, fully adaptive; all parameters.
$q \geq k \log(en/k)/128$		Sections 3–4, including the elementary GOE norm bound; all parameters.
$O(k \log(en/k)/\sqrt{\varepsilon})$ bound	upper	Appendix A and the preserved v3 note; charged scale estimation, exact-real model.
$\Omega(k \log(en/k)/\sqrt{\varepsilon})$ bound	lower	New sharp information lemma, proved warm start, imported SER spectral Theorem 6.1; condition (1.3).
Full RA-14 characterization		Not established for all simultaneous finite parameters.

10 Reproducibility and verification limits

The archive includes the source of this note, the original PDFs and ZIPs unchanged, explicit claim and dependency records, analytical-check code, and unit tests. New tests check exact rational spherical moments and constants, Gaussian covariance and likelihood identities, rank-one determinant identities, the covering perturbation factors, near-critical bin estimates, parameter absorption, and the scalar trace/gap implications. The prior upper-bound implementation and tests are retained separately and rerun without changing their assertions.

In the recorded run, all 20 new component tests and all 17 unchanged v3 tests passed. The 2,160 determinant-identity rows came from 90 prescribed-frame diagnostic sequences; their maximum recorded error was below $1.18 \cdot 10^{-14}$. These are mathematical component checks and floating-point implementation tests, not an enumeration of adaptive algorithms. In particular no numerical experiment proves the likelihood lemma, a universal lower bound, or the external spectral concentration theorem. The proofs above supply those arguments or identify the imported dependency. Floating-point computations are not the exact-real oracle model. Test counts and exit codes are recorded from the actual run in `results/verification.json`; failures are not removed from the logs.

To reproduce from the extracted directory:

```
python -m pip install -r requirements.txt
OPENBLAS_NUM_THREADS=1 python run_checks.py
python verify_manifest.py
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The manuscript is a generated research argument, not an independently certified proof. A reviewer should in particular check the change of measure under adaptive queries, the deterministic net-to-matrix implication, the rank-one determinant exponent, and the exact spectral-to-output transfer before treating the results as established literature.

A The dimension-sensitive upper bound

This appendix condenses the self-contained construction in the preceding package [3]. It also fixes the upper half of both matching results without depending on an unspecified spectral-extraction rule.

A.1 An oversampled Gaussian event

Let $n \geq \ell = 256r$ and $\Omega \in \mathbb{R}^{n \times \ell}$ be standard Gaussian. Split its rows in any fixed eigenbasis into Ω_1 with r rows and Ω_2 with $n - r$ rows. Except with probability at most $2e^{-9r} + 2e^{-4n}$,

$$\text{rank } \Omega_1 = r, \quad \left\| \Omega_2 \Omega_1^\dagger \right\|_2 \leq \sqrt{n/r}. \quad (\text{A.1})$$

Here is an elementary proof with the constants used in the algorithm. A $1/8$ -net of the unit sphere in $\mathbb{R}^a$ has at most 17^a points. For fixed unit $u \in \mathbb{R}^r$, $\left\| \Omega_1^\top u \right\|^2$ is chi-squared with ℓ degrees of freedom. Its moment-generating function gives tails $\exp[-\ell(a - 1 - \log a)/2]$ at squared-norm ratio a . At $a = 9/16, 25/16$, the exponent coefficient exceeds $1/20$. A net union bound and approximation give

$$\left\| \Omega_1 \right\|_2 \leq (10/7)\sqrt{\ell}, \quad \sigma_{\min}(\Omega_1) \geq (3/4 - (10/7)/8)\sqrt{\ell} = (4/7)\sqrt{\ell}$$

except with probability $2 \cdot 17^r e^{-\ell/20} \leq 2e^{-9r}$. For Ω_2 , nets on both spheres and scalar Gaussian tails show all net bilinear forms are at most $\sqrt{20n}$ in magnitude, except with probability $2 \cdot 17^{n-r+\ell} e^{-10n} \leq 2e^{-4n}$. Net approximation gives $\left\| \Omega_2 \right\|_2 \leq (4/3)\sqrt{20n}$. Dividing by the lower singular value yields

$$\left\| \Omega_2 \Omega_1^\dagger \right\|_2 \leq \frac{7\sqrt{20}}{48} \sqrt{n/r} < \sqrt{n/r},$$

which proves (A.1).

A.2 Estimating the unknown scale

Let $M = A^\top A \succeq 0$; each product with M costs two original queries. Set $r = k + 1$, $\delta = \varepsilon/4$, $\ell_1 = 256r$, and

$$d_1 = \left\lceil \frac{\log(4n/(r\delta))}{4\sqrt{\delta}} \right\rceil.$$

Form $\mathcal{K} = \text{range}[\Omega, M\Omega, \dots, M^{d_1}\Omega]$, with a Gaussian block of width ℓ_1 , and compute the r th eigenvalue μ of the compressed M . This costs at most $2\ell_1(d_1 + 1)$ original queries, including compression. Put $c = \lambda_r(M)$. If $c > 0$, the event (A.1) implies

$$(1 - \delta)c \leq \mu \leq c. \quad (\text{A.2})$$

To see the lower bound, put $t = (1 - \delta)c$ and use the analysis-only polynomial $\pi(x) = T_{d_1}(2x/t - 1)$. It has magnitude at most one on $[0, t]$, is positive and nondecreasing above c , and

$$\pi(c)^2 \geq \frac{1}{4} \exp(4d_1\sqrt{\delta}) \geq n/(r\delta),$$

using $\text{arcosh}((1 + \delta)/(1 - \delta)) = 2 \text{artanh } \sqrt{\delta} \geq 2\sqrt{\delta}$. In an eigenbasis partitioned after r entries, the graph

$$Y = \pi(M)\Omega\Omega_1^\dagger\pi(\Lambda_1)^{-1} = [I; F]$$

belongs to $\mathcal{K}$ and satisfies $Y^\top(M - tI)Y \succeq [\delta c - t \left\| \Omega_2 \Omega_1^\dagger \right\|_2^2 / \pi(c)^2] I \succeq 0$. Tail coordinates above t contribute nonnegatively; those at most t have $|\pi| \leq 1$. The min-max principle proves (A.2); its upper bound is Ritz interlacing. If $c = 0$, $M\Omega$ spans $\text{range } M$ almost surely. Conversely, if $c > 0$, the r th Ritz value is positive almost surely. Thus an exact test $\mu = 0$ gives a legitimate rank-promise branch.

A.3 A coarse polynomial surrogate

On $\mu > 0$ set $b = \mu/(1 - \delta)$, so $c \leq b \leq c/(1 - \delta)$. Put $\eta = \varepsilon/4$, $R^2 = 3n/k$, and

$$d_2 = \left\lceil \frac{\log(8R^2/\eta)}{2\sqrt{\eta}} \right\rceil, \quad H = T_{d_2}(2M/b - I).$$

Take a fresh Gaussian block of width $\ell_2 = 256k$. Compute $Q = \text{orth}(H\Omega)$, then HQ , and return the top k *right singular vectors* of $(HQ)^\top = Q^\top H$, completing orthonormally if needed. Each of the two polynomial applications costs at most $2\ell_2 d_2$ original queries via the Chebyshev recurrence.

In an eigenbasis of M , all but the first k entries of H have magnitude at most one. More generally, write a symmetric $H = \text{diag}(H_1, H_2)$ with H_1 of size k , $\|H_2\| \leq 1$, and let $L = \|\Omega_2 \Omega_1^\dagger\|_2$. The difference between H and its column-range approximation $H\Omega\Omega_1^\dagger[I \ 0]$ is

$$\begin{bmatrix} 0 & 0 \\ -H_2 \Omega_2 \Omega_1^\dagger & H_2 \end{bmatrix},$$

so $\|(I - QQ^\top)H\|_2^2 \leq 1 + L^2$. Also $\sigma_{k+1}(Q^\top H) \leq \sigma_{k+1}(H) \leq 1$. The prescribed right-singular-vector truncation therefore has $\|QQ^\top H(I - ZZ^\top)\|_2 \leq 1$. Adding the squared actions of the two orthogonal left ranges gives

$$\|H(I - ZZ^\top)\|_2^2 \leq 2 + L^2 \leq 2 + n/k \leq R^2.$$

No invertibility of H_1 is needed. Because H may be indefinite, substituting algebraic Ritz vectors would not be justified by this argument.

A.4 Returning to the spectral residual and counting queries

For every $z \geq 0$, the degree choice ensures

$$z \leq 1 + \eta + \frac{\eta}{R^2} T_{d_2}(2z - 1)^2. \quad (\text{A.3})$$

For $z \leq 1 + \eta$ this is immediate. For $z > 1 + \eta$, the function $T_d(2z - 1)^2/z$ is nondecreasing: if $y = \cosh u \geq 1$, then $yT'_d(y)/T_d(y) = d \coth u \tanh(du) \geq 1$, which makes its logarithmic derivative nonnegative. At $z = 1 + \eta$,

$$T_{d_2}(1 + 2\eta)^2 \geq \frac{1}{4} e^{2d_2 \sqrt{\eta}} \geq 2R^2/\eta \geq (1 + \eta)R^2/\eta,$$

using $\text{arcosh}(1 + 2\eta) = 2 \text{arsinh} \sqrt{\eta} \geq \sqrt{\eta}$. Thus (A.3) follows. Functional calculus yields $M \preceq (1 + \eta)bI + (\eta b/R^2)H^2$. Evaluating on unit vectors orthogonal to Z proves

$$\|A(I - ZZ^\top)\|_2^2 \leq (1 + 2\eta)b \leq \frac{1 + \varepsilon/2}{1 - \varepsilon/4} \sigma_{k+1}(A)^2 \leq (1 + \varepsilon)^2 \sigma_{k+1}(A)^2.$$

The two Gaussian failures have sum at most $2e^{-9(k+1)} + 2e^{-9k} + 4e^{-4n} < 0.001$ on the randomized branch $n \geq 256(k + 1)$. The algorithm uses the exact n -column recovery branch if the sketch width exceeds n or if that branch is cheaper than

$$B = 2\ell_1(d_1 + 1) + 4\ell_2 d_2 \leq 3072k + 1536 \frac{k}{\sqrt{\varepsilon}} \log \frac{96(n/k)}{\varepsilon}.$$

If $\varepsilon^{-1/2} \geq n/k$, the n cap already bounds the desired expression. Otherwise $\log(1/\varepsilon) < 2 \log(n/k)$, giving $B \leq 12000k\varepsilon^{-1/2} \log(en/k)$. The sketch-width fallback also satisfies this bound because $n < 256(k + 1) \leq 512k$. This proves the upper bound in (1.2).

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